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Development of a Monte Carlo track structure code for low-energy protons in water

S. UEHARA[†], L. H. TOBUREN[‡] and H. NIKJOO^{§*}

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Abstract.

Purpose: The development of a new generation of Monte Carlo track structure code is described, which simulates full slowing down of low-energy proton history tracks (*lephist*) in the range 1 keV–1 MeV in water.

Material and methods: All primary protons are followed down to 1 keV and all electrons to 1 eV. All primary interactions, including elastic scattering, ionization, excitation and charge exchange processes by protons and neutral hydrogen were taken into account. Cross-sections for proton and hydrogen impact were obtained from experimental data for water. Where data were lacking, the existing experimental data were fitted and extrapolated. The tracks of secondary electrons were generated using the electron track code *kurbuc*. The cross-sections and the energy transfer data were individually evaluated for the principal interactions induced by protons and hydrogen atoms in water. The analysis starts with the published cross-section data for water using a semi-empirical model including contributions from the neutral hydrogen atoms. For excitation cross-sections, the original Miller–Green analytical formula was used. For ionization by neutral hydrogen atoms, the same energy spectrum was assumed for secondary electrons as for protons. The total cross-sections were taken from the experiment of Blorizadeh and Rudd (1986b, c). For the stripping of charge by neutral hydrogen the data of Toburen *et al.* (1968) were used.

Results: Data are presented on total and differential elastic cross-sections as a function of energy and scattering angle respectively; single and double differential cross-sections for secondary electrons ejected by various energy proton impact; total cross-sections due to proton and hydrogen impact on water; stopping power cross-sections; and fraction of stopping power for water for protons as a functions of proton energy.

Conclusions: Tracks were analysed to provide confirmation on the reliability of the code and information on physical quantities, such as range, *W*, restricted stopping power, radial dose profiles and some microdosimetric parameters. Model calculations show good agreement with the experimental and calculated data.

1. Introduction

Over the past three decades many Monte Carlo track structure codes have been written to simulate the passage of charged particles in environments of

interest to biological, medical and other practical applications. As knowledge of the physics of interaction of charged particle in matter and speed of computers has increased, so has the range of application of these codes. For a comprehensive list, see Nikjoo *et al.* (1998). These computer simulation codes have mainly been used to study the nature of biological damage induced by ionizing radiation and have contributed significantly to the understanding of the mechanism of radiation effects in DNA and cells (Nikjoo *et al.* 1998, 1999).

All the ion codes (proton, α and heavy ions) including *moca14* (Wilson and Paretzke 1981), *orec* (Turner *et al.* 1983), *delta* (Zaider *et al.* 1983), *stbrgen* (Chatterjee and Holley 1993), *trion* (Lappa *et al.* 1993), *tracpro* (Cobut *et al.* 1998), *pits* (Wilson and Nikjoo 1999, Emfietzoglou *et al.* 2000) have a common feature in that they all simulate track segments and the effective energy range is limited to energies $> 0.3 \text{ MeV}^{-1}$ because of lack of cross-sections for many of the important interaction processes (Wilson and Nikjoo 1999).

Recent developments include calculation of theoretical cross-sections for the first time for liquid water $< 0.3 \text{ MeV}^{-1}$ by Dingfelder *et al.* (2000) and Uehara *et al.* (2000). These reports produce theoretical stopping cross-sections in good agreement with the published tabulated data. In this report, experimental cross-sections have been used for water vapour because experimental data for liquid water are not available.

This paper describes a new model for the development and simulation of full slowing down and extension of the proton track-segment simulation to lower energies $< 0.3 \text{ MeV}^{-1}$ in water. For track structure calculations, accurate treatment of all interactions are needed. All major interactions, including elastic scattering, excitation and ionization by dressed ions and neutrals, and charge-exchange processes which play an important role in the energy degradation of ions at these energies, are evaluated based on various experimental and theoretical data. The reliability of cross-sections and energy transfers are examined by analytical calculations of the macroscopic stopping

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cross-sections. More detailed descriptions for total cross-sections for ionization, excitation and charge exchanges, and the electronic stopping cross-sections are given in Uehara *et al.* (2000). The code *lephist* generates proton tracks in the energy range 1 keV to 1 MeV. Tracks of secondary electrons ejected by proton and neutral hydrogen were traced using the electron code *kurbuc* (Uehara *et al.* 1993). Various results such as range, W and the microdosimetric quantities derived by analysis of proton track are presented and discussed.

2. Interactions of protons and electrons at low energy

2.1. Proton interactions

As charged particles pass through matter they lose energy primarily through collisions with bound electrons. Ionization cross-sections at all projectile and secondary electron energies are needed to follow the history of an incident particle and its products, covering all ranges of energy transferred in individual collisions. For fast protons, the majority of energy is transferred in ionizing collisions, resulting in energetic free electrons and the potential energy of residual ions. Ionizing collisions involving bare ion projectiles, unless they are very slow, can be treated by using a classical description for the trajectory of the projectile with a quantal description of the target. ICRU (1996) provides an excellent review of the availability and the features of electron spectra originating from ion collisions with atoms and molecules.

When fast protons slow down, interactions involving electron capture and loss by the moving protons become an increasingly important component of the energy loss process. For incident protons with speeds comparable with or lower than the orbital speeds of the bound electrons in the target, the capture of electrons from the target is a significant mechanism that ionizes the target but does not produce secondary electrons. If the projectile is a neutral atom or a dressed ion (i.e. an ion bearing one or more electrons), a collision can eject electrons from either the projectile or the target. For a fast projectile, one can consider the projectile as a simple collection of a nucleus and electrons travelling together at the same speed. After the charge transfer process the neutral hydrogen atom becomes either stripped or ionizes a water molecule. In the first case, a proton and a possibly excited water molecule remains, an electron is ejected in forward direction with nearly the same velocity as the proton. In the latter case, the neutral hydrogen, an ionized water molecule and an ejected electron with a similar distribution of electron energy

and ejection angle as for protons result. Toburen (1998) gives a comprehensive review of the basic data of interactions for low-energy ions with various medium focussing on ionization and charge-transfer.

In addition to information on ionization and other energy-loss processes, excitation cross-sections for both protons and neutral hydrogen were taken into account in this work. There are no experiments on excitation cross-sections for proton impact for water. The proton cross-sections obtained by scaling of the electron excitation cross-sections are plausible for high-energy protons > 500 keV (Dingfelder *et al.* 2000). To cover the lower energy regions, the semi-empirical model developed by Miller and Green (1973) was adopted, which is based on the electron impact excitation.

Generally, high-energy protons proceed along a straight-line trajectory. However, the effects of elastic nuclear collisions will also become more prominent in the low energy region. Elastic collisions transfer little or no energy but can have a significant effect on the spatial character of the track structure at very low energies. Also, the nuclear energy loss cannot be neglected < 10 keV. It is assumed that the direction of proton track is not affected by events such as ionization, excitation and charge transfer for all energy regions, but is dominated only by the scattering angle in the laboratory system calculated by the theory of nuclear elastic scattering.

Table 1 lists the primary interactions of low-energy proton and neutral hydrogen with the stopping medium (water). The interactions included in further consideration are (1) nuclear elastic scattering, (2) target ionization and target excitation and (3) charge transfer consisting of electron capture and electron loss. These data are available over most of the energy range of interest and where data were lacking the existing data were fitted and extrapolated. Charge transfer can also be accompanied by simultaneous target ionization for both electron loss and electron capture processes as shown in the last two interactions in the table 1. However, there is insufficient information on the cross-sections for these interactions to be able to be included in these calculations.

2.2. Electron interactions

The transport of all secondary electrons is explicitly followed to 1 eV using the Monte Carlo electron code *kurbuc* (Uehara *et al.* 1993). The cross-sections considered in the code were electron impact ionization and excitation, and elastic scattering. The elastic scattering > 1 keV was calculated using the Rutherford formula taking into account the screening parameter given by Moliere (1948). The effective

Table 1. Interactions for low-energy proton and neutral hydrogen with water.

Interactions	Proton	Neutral hydrogen
Elastic scattering	$p + \text{H}_2\text{O} \rightarrow p' + \text{H}_2\text{O}$	$\text{H} + \text{H}_2\text{O} \rightarrow \text{H}' + \text{H}_2\text{O}$
Target ionization	$p + \text{H}_2\text{O} \rightarrow p' + \text{H}_2\text{O}^+ + e$	$\text{H} + \text{H}_2\text{O} \rightarrow \text{H}' + \text{H}_2\text{O}^+ + e$
Target excitation	$p + \text{H}_2\text{O} \rightarrow p' + \text{H}_2\text{O}^\bullet$	$\text{H} + \text{H}_2\text{O} \rightarrow \text{H}' + \text{H}_2\text{O}^\bullet$
Electron capture	$p + \text{H}_2\text{O} \rightarrow \text{H}' + \text{H}_2\text{O}^+$	—
Electron loss	—	$\text{H} + \text{H}_2\text{O} \rightarrow p' + e + \text{H}_2\text{O}$
Electron capture and target ionization	$p + \text{H}_2\text{O} \rightarrow \text{H}' + \text{H}_2\text{O}^{2+} + e$	—
Electron loss and target ionization	—	$\text{H} + \text{H}_2\text{O} \rightarrow p' + e + \text{H}_2\text{O}^+ + e$

The symbol (') for p and H at the right-hand side means that the projectiles lose a small part of their energies in the interactions.

atomic number and the screening parameter were chosen to fit the experimental data. Angular distribution for elastic scattering < 1 keV were obtained by direct sampling of various experimental data. A least-squares fitted total cross-section for elastic scattering, including both experimental and theoretical data, was used in place of cross-sections used in the original version (Uehara *et al.* 1999). The fitted data includes various sets of experimental data (Danjo and Nishimura 1985, Katase *et al.* 1986, Shyn and Cho 1987, Hayashi 1989) and one theoretical data set (NIST 1996). Inelastic cross-sections for ionizations and excitations were compiled from different sources for low- and high-energy electrons. In the range 10 eV to 10 keV, total ionization cross-sections were obtained from the least-square fitting of various experimental data (Schutten *et al.* 1966, Olivero *et al.* 1972, Bolorizadeh and Rudd 1986a, Djuric *et al.* 1988, Hayashi 1989). Excitation cross-sections were obtained from the data of Paretzke (1988). The calculation of the secondary electron spectrum was carried out using the method of Seltzer based on the Weizsacker–Williams method (1988). Angular distributions of secondary electrons were calculated using a combination of the experimental data of Opal *et al.* (1972) and the kinematic relationship for low- and high-energy electrons.

The existing Monte Carlo electron track structure codes (Nikjoo *et al.* 1998) use a variety of theoretical models and assumptions in treating physical and chemical processes (Nikjoo *et al.* 1994). Inter-comparison and assessment of various experimental and theoretical approaches adopted in a number of Monte Carlo track structure codes were performed using the code *kurbuc* as an engine to observe the effect of the change of a set of cross-sections on the output (Uehara *et al.* 1999). It was concluded that most codes produce tracks in reasonable agreement with the macroscopic quantities such as total stopping power and total yield of strand breaks. However, differences were predicted in frequencies of clustering in tracks from different models.

3. Proton cross-sections

3.1. Elastic scattering

The interaction between the projectile and the target atom can be described in terms of a potential function

$$V(r) = (zZe^2/r)F_s(r/r_s) \quad (1)$$

The first factor in equation (1) is the Coulomb potential for two bare nuclei with charge ze and $\tilde{Z}e$. The factor $F_s(r/r_s)$ takes into account the screening by atomic electrons. The extent of screening is characterized by the screening length, r_s . A commonly used prescription for the screening length is

$$r_s = 0.88534r_B(z^{2/3} + Z^{2/3})^{-1/2} \quad (2)$$

where r_B is the Bohr radius and 0.88534 is a numerical constant from the Thomas–Fermi model of the atom. For protons, a screening function $F_s(r/r_s)$ is given by (ICRU 1993)

$$F_s(r/r_s) = 0.10e^{-6r/r_s} + 0.55e^{-1.2r/r_s} + 0.35e^{-0.3r/r_s} \quad (3)$$

For a particle scattered in a central potential $V(r)$, the deflection angle θ is obtained in a classical-mechanics trajectory calculation as a function of the impact parameter, p (Mott and Massey 1965)

$$\theta = \pi - 2 \int_{r_{\min}}^{\infty} \frac{p}{r^2 \sqrt{1 - V(r)/T_{\text{cm}} - p^2/r^2}} dr \quad (4)$$

where $r_{\min}$ is the distance of closest approach. The positive root of the following gives this value

$$r^2 - 2zZr_e \frac{m_e c^2}{\beta P_c} r F_s(r/r_s) - p^2 = 0 \quad (5)$$

in which

$$\beta = \sqrt{1 - \left(\frac{m_p c^2}{T + m_p c^2} \right)^2} \quad (6)$$

$$P_c = \sqrt{(T + m_p c^2)^2 - (m_p c^2)^2}$$

and $r_e = 2.81794 \times 10^{-13}$ cm, $m_e c^2$ = electron rest mass, $m_p c^2$ = proton rest mass and P is the momentum of proton. T_{cm} is the particle energy in the centre-of-mass system, given by

$$T_{\text{cm}} = \frac{T}{1 + m_p/M_t} \quad (7)$$

where T is the energy in the laboratory system and m_p and M_t are the masses of proton and target atom. Equation (4) was solved numerically using procedures given by Everhart *et al.* (1955) and Wilson and Haggmark (1977). The deflection angle θ was thereby obtained as a function of the impact parameter p . The boundary of ‘large-angle calculations’ and ‘small-angle calculations’ described by Everhart *et al.* was set at $\theta = 0.1\pi$ where smooth transitions are given.

The differential elastic scattering cross-section can be obtained by numerical differentiation of the curve of impact parameter versus deflection angle using a B-spline interpolation (Goldstein 1950)

$$\frac{d\sigma_{\text{el}}}{d\Omega} = - \frac{p}{\sin \theta} \frac{dp}{d\theta} \quad (8)$$

where θ and $d\sigma_{\text{el}}/d\Omega$ are functions of the impact parameter and the particle energy. The energy deposition ΔE in elastic scattering depends on both the scattering angle and the proton energy. This is given by the formula $W(\theta, T)$ according to ICRU (1993)

$$\Delta E = W(\theta, T) = 4T \frac{m_p M_t}{(m_p + M_t)^2} \sin^2 \frac{\theta}{2} \quad (9)$$

To reduce the divergence of the total cross-section, the cut-off angles, θ_{cut} , were set such as to limit the increase in the scattering probability at low scattering angles (ICRU 1978)

$$\theta_{\text{cut}} \approx \frac{1}{137} Z^{1/3} \frac{m_e c^2}{Pc} \quad (10)$$

Total elastic scattering cross-section for the proton energy T is calculated by

$$\sigma_{\text{el}}(T) = 2\pi \int_{\theta_{\text{cut}}}^{\pi} \frac{d\sigma_{\text{el}}}{d\Omega} \sin \theta d\theta \quad (11)$$

Scattering angle θ is randomly sampled by a table storage and look-up method in the Monte Carlo track code. The table was predigested by separated data preparation routines based on the direct sampling method using differential cross-sections (equation 8). Figure 1 shows examples of 10^5 times randomly sampled angular distributions of elastic scattering at small angles for the proton energies of 1, 10, 100 keV and 1 MeV. Figure 2 shows θ_{cut}

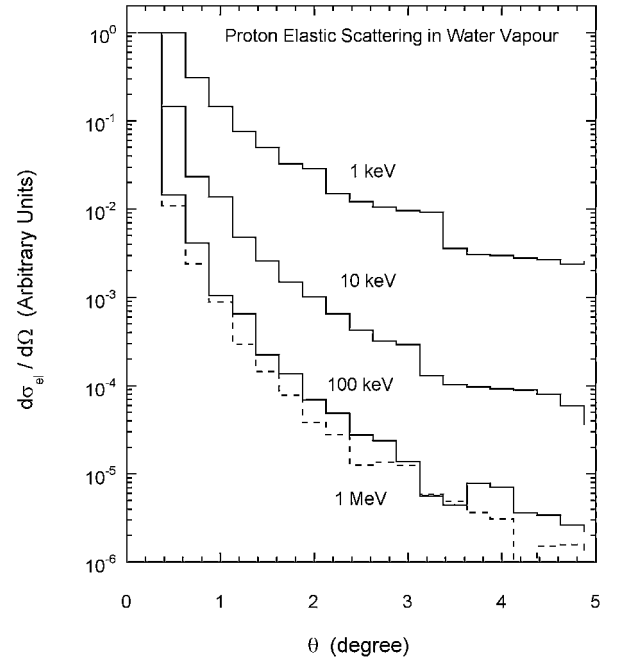


Figure 1. Differential cross-sections for proton elastic scattering in water vapour ($d\sigma_{\text{el}}/d\Omega$ in arbitrary units) as a function of scattering angle θ . The step lines are the results obtained from sampling 10^5 times using the calculated data from equation (8).

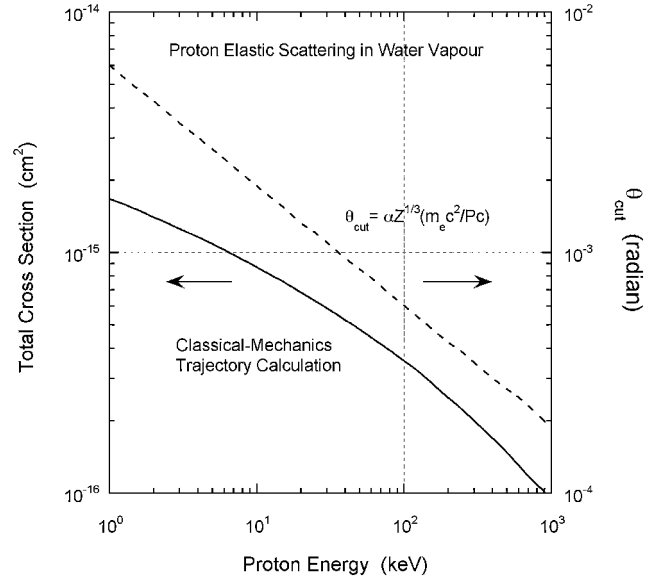


Figure 2. Total cross-section evaluated by the classical-mechanics trajectory calculation (solid line) and the cut-off angle θ_{cut} at the small angle (dashed line) for proton elastic scattering in water vapour.

(equation 10) and the total cross-section (equation 11) of proton elastic scattering in water vapour as a function of T . It was assumed that the elastic scattering cross-sections for neutral hydrogen (H^0) to be equal to that for protons.

3.2. Ionization

3.2.1. *Total cross-sections.* Total ionization cross-sections for protons $\sigma_{ip}(T)$ were determined by the least-square fit to the experimental data given by Rudd *et al.* (1985), in which the six data sets measured under the different experimental conditions are provided. These six data sets fall into the two groups, two sets are larger in magnitude than the other four. Only the four sets belonging to the smaller group consisting of ‘Low-energy accelerator’, ‘early Van de Graaff data’, ‘early UNL data’ and ‘late UNL data’ gave consistent agreement with published stopping cross-sections in the high energy regions, and were therefore used for subsequent calculations. The four sets were fitted by power functions at the lower and the higher energy regions

$$\sigma_{ip}(T) = 10^\alpha \quad (12)$$

$$\begin{aligned} \alpha = & -0.24653 + 0.68546 \log T + 0.11287(\log T)^2 \\ & - 0.11194(\log T)^3 \text{ for } 1 \leq T \leq 100 \text{ keV} \\ \alpha = & -3.96257 + 6.08376 \log T - 2.46783(\log T)^2 \\ & + 0.29425(\log T)^3 \text{ for } 100 \text{ keV} < T \leq 1 \text{ MeV} \end{aligned}$$

where log is to base 10. The cross-sections from the fits have units of 10^{-16} cm^2 with T in keV.

The total electron production cross-section for neutral hydrogen impact, σ_{iH} , can be obtained by twice integrating the double-differential cross-section (DDCS) measured by Bolorizadeh and Rudd (1986c). According to their notation, the cross-section obtained in this way is represented by $\sigma_{\cdot}$. However, $\sigma_{\cdot}$ includes contributions from electrons ejected from the target and from the projectile. The electrons from the projectile appear preferentially in the forward direction and with essentially the velocity of the now ionized projectile. This phenomenon was called electron loss to the continuum (ELC) (Bolorizadeh and Rudd 1986c). The cross-section for ionization of the target molecule by neutral hydrogen, σ_{iH} , was obtained by subtracting the electron loss cross-section $\sigma_{\cdot}(\text{ELC})$ from $\sigma_{\cdot}$. $\sigma_{\cdot}$ and $\sigma_{\cdot}(\text{ELC})$ were taken from table I of Bolorizadeh and Rudd (1986c) for the energy range between 20 and 150 keV. σ_{iH} was then fitted by the function

$$\begin{aligned} \sigma_{iH}(T) = & 10^\alpha \text{ for } 20 \leq T \leq 150 \text{ keV} \\ \alpha = & -0.95852 + 2.0153 \log T \\ & - 0.56507(\log T)^2 \end{aligned} \quad (13)$$

The cross-section from the fit has units of 10^{-16} cm^2 with T in keV.

As there are no experimental data for ionization by neutral hydrogen for the energy regions $< 20 \text{ keV}$ or $> 150 \text{ keV}$, we assumed similarity with proton impact ionization. Cross-sections for neutral

hydrogen ionization were approximated by scaling proton ionization to give smooth transitions at 20 and 150 keV. The results are

$$\sigma_{iH}(T) = 1.3097 \sigma_{ip}(T) \text{ for } T < 20 \text{ keV} \quad (14)$$

and

$$\sigma_{iH}(T) = 1.3454 \sigma_{ip}(T) \text{ for } T > 150 \text{ keV} \quad (15)$$

3.2.2. *Energy spectra for secondary electrons.* Energy spectra of secondary electrons ejected by proton impact were calculated using the empirical model given by Rudd (1988)

$$\frac{d\sigma_i(\varepsilon)}{d\varepsilon} = \frac{S}{B} \frac{F_1 + F_2 w}{(1+w)^3 \{1 + \exp[\alpha(w - w_c)/v]\}} \quad (16)$$

where ε = electron energy, B = binding energy for each orbital, $\lambda = m_p/m_e = 1836$, $a_0 = 0.529 \times 10^{-8} \text{ cm}$, N = electron number in orbital, $R = 13.6 \text{ eV}$, $w = \varepsilon/B$, $v = (T/\lambda B)^{1/2}$, $w_c = 4v^2 - 2v - R/4B$, and $S = 4\pi a_0^2 N(R/B)^2$

$$\begin{aligned} F_1(v) &= L_1 + H_1 \\ L_1 &= C_1 v^{D_1} / (1 + E_1 v^{D_1 + 4}) \\ H_1 &= A_1 \ln(1 + v^2) / (v^2 + B_1 / v^2) \\ F_2(v) &= L_2 H_2 / (L_2 + H_2) \\ L_2 &= C_2 v^{D_2} \\ H_2 &= A_2 / v^2 + B_2 / v^4 \end{aligned} \quad (17)$$

The fitting parameters for water vapour are given by Rudd *et al.* (1992)

$$\begin{aligned} A_1 &= 0.97 & B_1 &= 82 \\ A_2 &= 1.04 & B_2 &= 17.3 \\ C_1 &= 0.40 & D_1 &= -0.30 \\ C_2 &= 0.76 & D_2 &= 0.04 \\ E_1 &= 0.38 & \alpha &= 0.64 \end{aligned}$$

This model gives the differential cross-sections for secondary electron production by proton impact, and also the partial ionization cross-sections for each orbital. The boundary energy for energetic secondary electrons generated by ionization was determined as 1 eV after the measurement condition of Bolorizadeh and Rudd (1986b). If electron energy $< 1 \text{ eV}$, then the energy deposition is $\Delta E = B_i + \varepsilon$ for the i th orbital ionization.

The secondary electron energy is also sampled by a storage and look-up methods in the Monte Carlo track code. Figure 3 presents the randomly sampled energy spectra of ejected electrons at $T = 20, 50$ and 150 keV and 1 MeV in comparison with the experimental data of Bolorizadeh and Rudd (1986b)

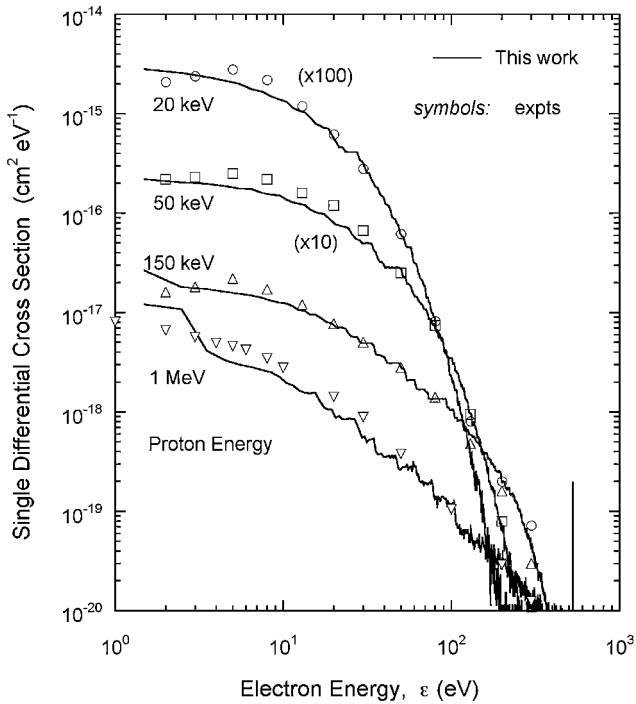


Figure 3. Randomly sampled energy spectra for secondary electrons ejected by proton impact in comparison with experimental data taken from Bolorizadeh and Rudd (1986b) for $T=20, 50$ and 150 keV and from Wilson *et al.* (1984) for $T=1$ MeV. The lines are the results obtained from sampling 10^5 times using calculated data from the Rudd model (equations 16 and 17).

and Wilson *et al.* (1984). Because the experimental data include the contribution from all orbitals, calculations took into account the relative intensity of the partial ionization cross-sections. Data obtained from sampling 10^7 times were normalized for the overall fitting to the experimental values. Good agreement shows the appropriateness of the Rudd's model. It was assumed that the spectra of electrons ejected from the target molecule by neutral hydrogen impact to be equal to that for protons.

3.2.3. Angular distributions for secondary electrons. Experimental angular distributions of electrons ejected from water by proton impact are given graphically by Bolorizadeh and Rudd (1986b) at the proton energies of 15 and 100 keV and by Toburen and Wilson (1977) at the proton energies of 0.3, 0.5 and 1 MeV. For random sampling of the angular distributions of ejected electrons, we directly used the experimental data for various electron energies at $T=15, 100, 300, 500$ keV and 1 MeV, and interpolated for the intermediate energies of electrons and protons. Figure 4 presents a comparison of the angular distributions of ejected electrons having various energies between randomly sampled and experiments

at $T=100$ keV (left) and $T=500$ keV (right) in which ε is the electron energy in eV units. Sampled values obtained from sampling 10^5 times were normalized for overall fitting to the experimental values.

ICRU (1996) recommended the semi-empirical formula for the DDCE called Hansen–Kochbach–Stolterfoht (HKS) model, which appears to be very promising. Hansen and Kochbach (1989) derived a simple expression for the double differential electron emission probabilities as a function of the impact parameter. The derivation was based on various model assumptions including the semi-classical approximation. The initial electron state is described by the screened hydrogenic wave function and a plane wave was chosen to represent the outgoing electron. The readily calculable form for numerical evaluation of the DDCE is given by equation (4.34) in ICRU (1996), which was obtained by integrating the probability function over the impact parameter.

$$\frac{d^2\sigma_i(\varepsilon, \theta)}{d\varepsilon d\Omega} = \frac{16a_0^2 Z_1^2}{3\pi R v_1^2 \alpha_c^3} \left[\frac{\alpha_c^4}{\alpha_c^4 + (K_m - k_t \cos\theta)^2} \right] \quad (18)$$

$$k_c = \sqrt{k^2 + \frac{2\alpha^2}{\ln(2v_1^2/\alpha^2)}}$$

$$k_1 = \sqrt{k^2 + 0.2\alpha^2} \sqrt{v_1/\alpha}$$

$$\alpha_c = \alpha \left(1 + 0.7 \frac{v_1^2}{v_1^2 + k^2} \right)$$

where Z_1 is the charge and $v_1 = (T/\lambda R)^{1/2}$ the velocity of the projectile, $K_m = (\alpha^2 + k^2)/2v_1$ is the minimum momentum transfer. k_c and k_t are small modifications of the quantity $k = (\varepsilon/R)^{1/2}$, which is the momentum of the outgoing electron. α_c is a modification of the mean initial momentum parameter $\alpha = (B/R)^{1/2}$. This equation is for a single-target electron and must be summed over all the electrons in the target.

The DDCE calculated using equation (18) are shown in figure 4 (broken lines) in comparison with the experimental data. The right panel for 500 keV protons shows a fairly good agreement between the HKS model and the experimental angular distributions for the ejected electrons with various energies. However, the reproducibility for 100 keV protons is unsatisfactory. As the theoretical and model fitting do not provide a satisfactory solution, it is then realistic, to use random sampling of angular distributions of ejected electrons from the experimental and interpolated data.

3.3. Excitation

Excitation cross-sections and mean excitation energy loss by proton and hydrogen impact on water

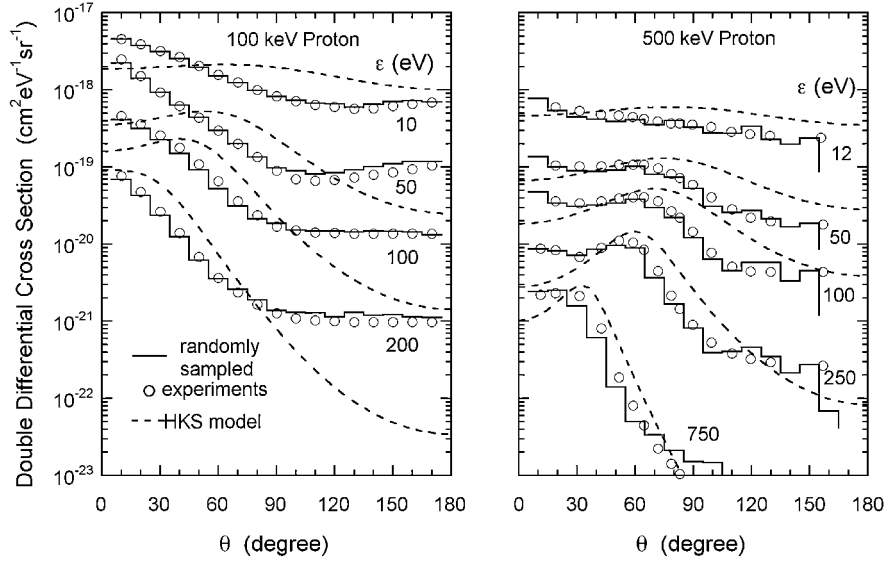


Figure 4. Comparison of angular distributions of secondary electrons for various energies ε (eV) ejected by proton impact. The circles are experimental data of Bolorizadeh and Rudd (1986b) for $T=100$ keV (left) and those of Toburen and Wilson (1977) for 500 keV (right). The solid lines are the results obtained from sampling 10^5 times from these data using the direct sampling method. The broken lines are the calculated DDCS by means of the HKS model (ICRU 1996).

molecules were treated based on the formula given by Miller and Green (1973). They assumed an analytical function for each excited level of the form

$$\sigma_c(T) = \frac{\sigma_0 (\zeta a)^\Omega (T - W)^\nu}{\mathcal{J}^{\Omega+\nu} + T^{\Omega+\nu}} \quad (19)$$

where $\sigma_0 = 10^{-16} \text{ cm}^2$ and $Z=10$; W , a , $\mathcal{J}$, Ω and ν for 28 excitation states are given in table III of Miller and Green (1973). In the work of Miller and Green, cross-sections for excitation by neutral hydrogen were calculated using the same values as for protons except the parameter a of equation (19) was assumed to be three-quarters of the proton value. They noted that their excitation cross-sections were only rough estimates proposed in the absence of detailed theoretical or experimental data. They further noted that the position and height of the maximum in the cross-section could easily be in error by a factor of two and that the nature of the low energy fall-off was speculative. The mean excitation energy W_{av} was calculated using the relationship $\Sigma W_j \sigma_{ej}(T) / \Sigma \sigma_{ej}(T)$, where W_j is the excitation threshold energy and $\sigma_{ej}(T)$ is the excitation cross-section of form 19 above for state j given by Miller and Green (1973). The mean energy transfer for excitation increases slowly from ~ 12 eV at 1 keV to 14 eV for the highest particle energies.

3.4. Charge exchanges

3.4.1. *Total cross-sections.* As low-energy charged particles pass through matter they undergo electron

capture and loss events. Total cross-sections for electron capture $\sigma_{10}(T)$ and electron loss $\sigma_{01}(T)$ for protons and neutral hydrogen, respectively, were evaluated using the analytical functions developed by Miller and Green (1973) and fitted to the experimental data of Toburen *et al.* (1968) and Dagnac *et al.* (1970). Recent experimental cross-sections for electron capture (Lindsay *et al.* 1997) are in good agreement with those of Dagnac *et al.* (1970). Electron capture is fitted by the equation as

$$\sigma_{10}(T) = \sigma_0 \left[\left(\frac{T}{A} \right)^p + \left(\frac{T}{B} \right)^q + \left(\frac{T}{C} \right)^r \right]^{-1} \quad (20)$$

where $\sigma_0 = 2 \times 10^{-15} \text{ cm}^2$, $A=44.1 \text{ keV}$, $B=6.0 \text{ keV}$, $C=1.5 \text{ keV}$, $p=3.52$, $q=0.667$ and $r=-0.027$. Electron loss $\sigma_{01}(T)$ is fitted by the same formula as equation (19) but the parameters: $\sigma_0 = 10^{-16} \text{ cm}^2$, $Z=10$, $a=79.3 \text{ keV}$, $W=12.6 \text{ eV}$, $\mathcal{J}=27.7 \text{ keV}$, $\Omega=0.652$ and $\nu=0.943$.

3.4.2. *Energy transfer model.* In the calculations of stopping cross-sections for low-energy protons, three energy-loss components were taken into account for energy transfer in the electron capture process (Uehara *et al.* 2000). These include: (1) energy to remove the electron from the target molecule, (2) energy to provide translational velocity for the electron to move at the speed of the proton and (3) energy released from capture into the bound state of the proton. Therefore, the net energy lost by the proton in capturing an electron is given by $Q_{10} = T/\lambda + B_{H_2O} - B_H$, where $B_H = 13.6 \text{ eV}$. The energy

lost by the neutral hydrogen atom in the electron-loss process is simply the binding energy of the projectile; $Q_{01} = B_H$. The kinetic energy of the electron resulting from the electron-loss process was provided for in the capture process.

From the track structure view point, it seems natural to regard the stripped electron as the real secondary electron. In fact, the stripped electron is observed in the spectrum of secondary electrons produced in neutral hydrogen – H_2O interactions as a peak centred at $\varepsilon = T/\lambda$ (Wilson and Toburen 1973, Bolorizadeh and Rudd 1986c). From such considerations, we adopted another energy deposition model in charge exchange processes in which while keeping the energy balance the same as listed above, the energy deposited ΔE in electron capture (Q_{j0}) was assumed to be equal to the average binding energy of water molecule

$$\Delta E = B_{av} \quad (21)$$

where B_{av} is the weighted mean of the binding energies $\sum p_i B_i$, in which p_i is the fractional ionization cross-sections for four outer-shells having the binding energy B_i calculated using the Rudd's model (equations 16 and 17). In the electron-loss process (Q_{01}) an electron is generated with the energy

$$\varepsilon = T/\lambda \quad (22)$$

and emitted with the same direction as the neutral hydrogen. The present treatment is equivalent to the model of Miller and Green (1973) in which the net effect of the cycle was the production of an ion pair and therefore total energy lost was the ionization threshold of H_2O^+ plus the kinetic energy of the stripped electron. All data regarding this electron, the kinetic energy T/λ , the direction and the position are documented in the output file together with the secondary electrons due to ionization by protons and neutral hydrogen.

3.5. Stopping cross-sections

Figure 5 shows the total cross-sections for all interactions induced by protons (left panel) and neutral hydrogen (right panel) in the energy range between 1 keV and 1 MeV. The reliability of basic cross-section data appropriate to the macroscopic stopping of low-energy protons in water was examined. Analytical calculations of nuclear and electronic stopping cross-sections for low-energy protons were performed using the cross-sections and mean energy loss per collision data.

The nuclear stopping cross-section at the proton energy T is given by

$$L_n(T) = 2\pi \int_{\theta_{cut}}^{\pi} \sigma_d(\theta) W(\theta, T) \sin \theta d\theta \quad (23)$$

where the energy transfer to the recoiling atom, $W(\theta, T)$, is given by equation (9). The electronic stopping cross-section for protons is defined by

$$L_e(T) = \sum_j f_p Q_j \sigma_j + \sum_k f_H Q_k \sigma_k \quad (24)$$

where j is interactions leading to ionization, excitation and electron capture by proton impact, and k is ionization, excitation and electron loss by neutral hydrogen impact. σ is the total cross-sections for processes j and k for proton and neutral hydrogen interactions respectively. Q is the mean energy transfer in each interaction and f_p and f_H are the equilibrium charge fractions for protons and neutral hydrogen, respectively. More detailed descriptions for calculations of electronic stopping cross-sections are given in Uehara *et al.* (2000).

Figure 6 shows the calculated nuclear and electronic stopping cross-sections of water vapour in comparison with the published data. The nuclear stopping cross-sections agreed well with the ICRU data (1993). The electronic stopping cross-sections are consistent with published data to within $\sim 20\%$ for energies < 1 MeV. It is difficult to identify reasons for discrepancies because of lack of experimental cross-sections for many of the interactions especially at lower energies < 10 keV.

4. Results and discussion

4.1. Ranges and W values

Figure 7 demonstrate a typical track segment of a 100 keV proton and three examples of full slowing down tracks of 1, 10 and 100 keV protons generated at the origin and along the z -axis. All primary protons were followed down to a cut-off energy at which point the residual energy was deposited at a random short distance. The cut-off energy was chosen to produce correct W and range values. The cut-off was set at 500 eV for 1 keV and at 1 keV at proton energies > 5 keV. The cut-off values in between 1 and 5 keV were interpolated. All electron interactions were followed down to 1 eV. In panel (a) the proton path together with those of the secondary electrons were followed up to $z = 100$ nm where the proton energy degrades down to 89 keV. The energy deposition points are represented by the dots or marks whose numbers of 71 for p -elastic, 110 for p -ionization, 51 for p -excitation, 28 for electron capture, 42 for H^0 -elastic, 55 for H^0 -ionization, 22 for

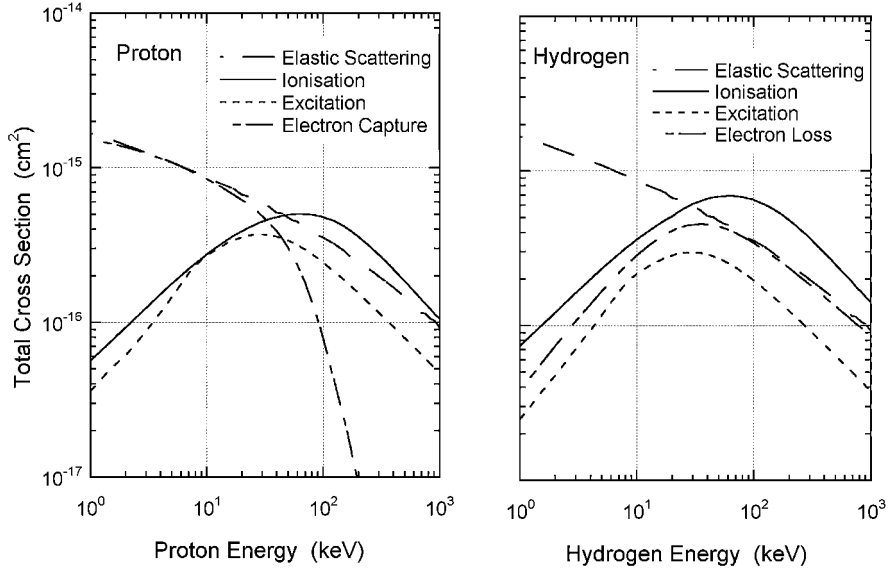


Figure 5. Total cross-sections due to proton (left) and hydrogen (right) impact on water vapour. The curve for elastic scattering was assumed to be the same between protons and neutral hydrogen.

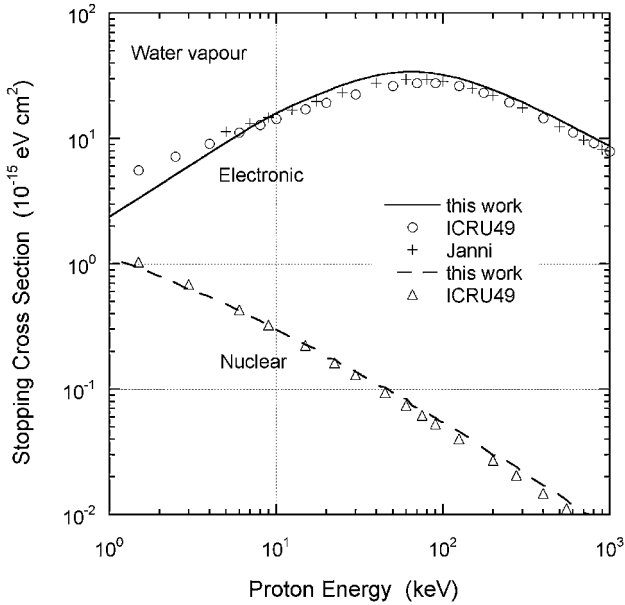


Figure 6. Comparison of the nuclear and the electronic stopping cross-sections of water vapour as a function of the impact energy of protons. Present calculations: nuclear (broken line) and electronic (solid line). Tabulations: ICRU (1993) and Janni (1982).

H^0 -excitation, 28 for electron loss and 776 (number of events) for secondary electrons are also included. Electron capture (open circles) and electron loss (triangles) appear alternately along the projectile path. The solid circles represent the energy deposited points by the secondary electrons. The small dots are the ionization potential, excitation and elastic scattering of protons and hydrogen. For electrons with a

relatively large energy, Bethe and Ashkin (1953) gave the classical law for the emission angle as $\cos\theta = (\varepsilon/4T)^{1/2}$. One can see many electron tracks are approximately perpendicular to the proton track. Panels (b–d) show the slowing tracks for 1, 10 and 100 keV protons.

To confirm the reliability of the simulated tracks and the code, calculations of macroscopic quantities such as the proton ranges and the W values were performed. Figure 8 shows the calculated proton ranges in water vapour (density = 1 g cm^{-3}) as a function of the primary energy of protons. The present data agreed well with the ICRU data (1993) $< 30 \text{ keV}$ and the tabulated data of Janni (1982) within 10% deviation $> 30 \text{ keV}$ while, the ICRU data for the energies $> 30 \text{ keV}$ were greater than the present data by 20%. This reflects the discrepancies of the electronic stopping cross-sections as shown in figure 6 between the present calculations and the ICRU data, in which the present data are greater than the ICRU data. Again the point needs to be emphasized that the discrepancies arise from lack of experimental data for confirmation of theoretical calculations. ICRU data have been calculated using the effective charge transfer model while we have used the experimental cross-sections and the mean energy transfer model (figure 5 and equation 24).

Figure 9 shows the calculated W values in comparison with published data (Miller and Green 1973, Stonell *et al.* 1993, Siebert *et al.* 1994, Grosswendt and Baek 1995). W values were derived from scoring the number of electron- H_2O^+ pairs due to ionization and charge exchange cycle by taking into account

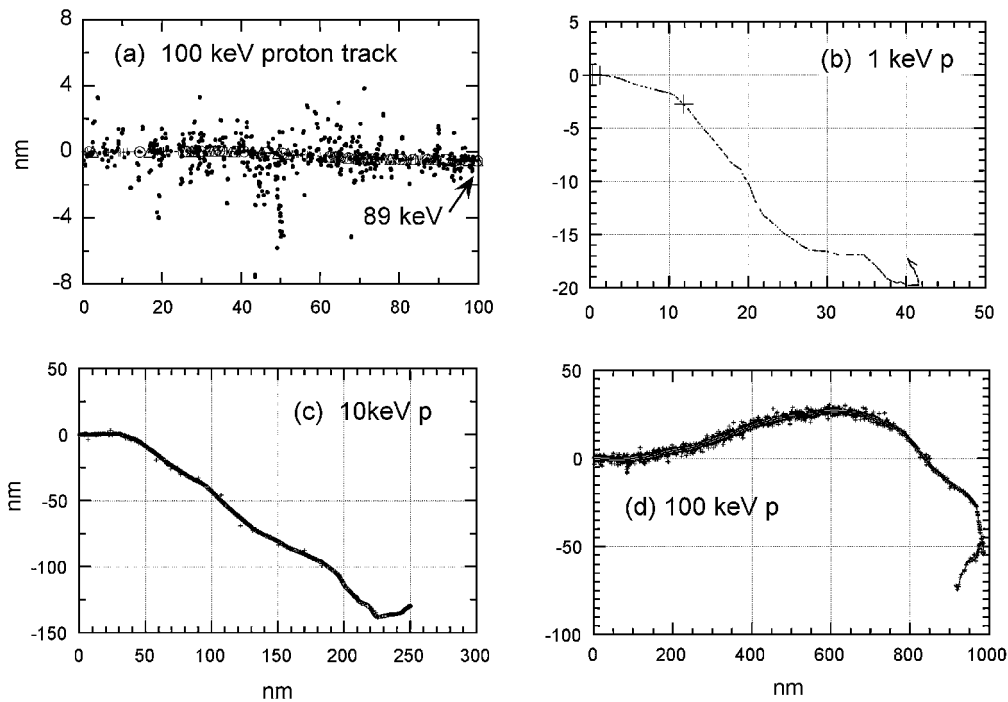


Figure 7. (a) Two-dimensional representation of part of the history of a track of a 100keV proton to 89keV. Dots or symbols represent the energy deposited at points where an interaction occurred. The evidence of the charge exchange cycle is shown by the symbols (circle) for electron capture and (triangle) for electron loss. (b–d) Tracks for 1, 10 and 100keV protons.

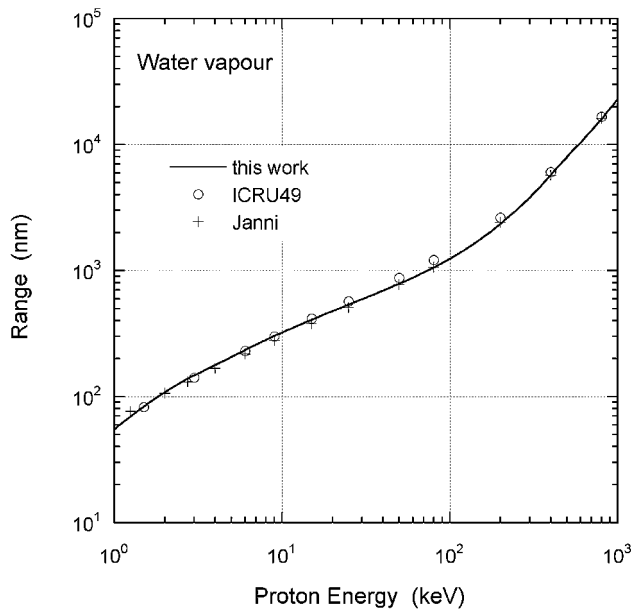


Figure 8. Calculated proton ranges as a function of the energy of protons in comparison with the data of ICRU (1993) and Janni (1982).

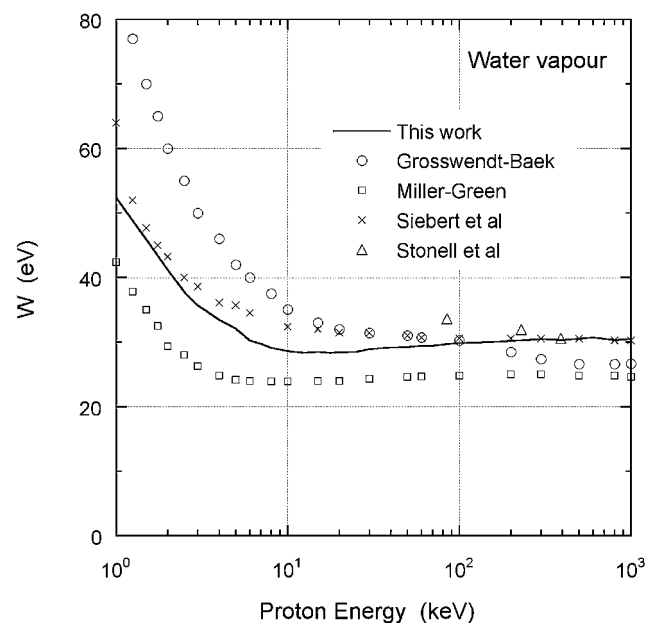


Figure 9. Calculated W values for protons as a function of the energy of protons in comparison with the data of Grosswendt and Baek (1995), Miller and Green (1973), Siebert *et al.* (1994) and Stonell *et al.* (1993). $W = 30.5$ eV at the high-energy limit was exactly the same as the experimental data of Siebert *et al.* (1994).

the ionization events by the energetic secondary electrons ($\epsilon > 12.62$ eV). The number of electron-ion pairs produced by the secondary electrons exceeds that of pairs produced by proton, neutral hydrogen and charge exchange cycle at ~ 500 keV, where the

number of electron induced pairs amounts to 8300 per proton. W values at a few keV is subject to considerable fluctuation because the number of pairs depends on the cut-off for the proton history. Here, we adjusted the cut-off to fit the experimental W and the ranges < 5 keV primary proton energy. Energy dependence of the W values shows a flat minimum ~ 10 keV, which is similar to the experimental data for protons in CO_2 and N_2 (Waibel and Willems 1992). W values of 30.5 eV at the high-energy limit was exactly the same as the experimentally determined asymptotic limit (Christophorou 1971) and the data evaluated from the analytical representation of $W(T)$ derived by Siebert *et al.* (1994) from the known experimental W of protons in methane-based tissue-equivalent gas.

4.2. Secondary electrons

Figure 10 shows the average and the maximum energies of secondary electrons as a function of the starting energy of the proton. The average energy here does not mean the average value for a single proton energy obtained from the equations (16) and (17). The present average energies were obtained by dividing the sum of energies of all electrons generated divided by the total number of electrons throughout the degradation of protons as low as 1 keV. The present values, e.g. 8.3 eV for 20 keV and 27 eV for 150 keV protons, include the contributions from electrons generated by the degraded proton energies and

therefore become smaller than the experimental data measured for a single proton energy. Bolorizadeh and Rudd (1986b) obtained the average energies of 13 and 37 eV at the single proton energies 20 and 150 keV respectively.

The maximum energy of secondary electrons depends only on the primary energy of projectile except for statistical fluctuations. The trial numbers were 100 and 5 for the proton energies of 100 keV and 1 MeV respectively. The deviation from the smooth curve can be seen at several energies. Assuming a head-on, linear collision between an incident proton and a free electron initially at rest, the kinematics gives the maximum energy of secondary electron as $4T/\lambda$ where T is the proton energy and $\lambda = 1836$. This formula is correct for the proton energies higher than several hundred keV as shown in figure 10 (short dashed line). The effect of the orbital binding energy taken into account in the Rudd's model adopted here becomes significant for the proton energies lower than 300 keV, where the deviation from $4T/\lambda$ increases as the proton energy decreases. The few Auger electrons having the energy of 527.1 (= 539.7 – 12.6) eV ejected when inner-shell ionization occurs were excluded in the analysis of the maximum energy.

5. Model calculations

5.1. Fractional stopping powers

Total and fractional stopping powers can be evaluated analytically by the equations 23 and 24. On the other hand, it is possible to obtain approximate stopping powers using the track structure data. In accordance with the definition of stopping power a short track length, such as that shown in figure 7a, was selected tentatively as 5% of the full range. Figure 11 shows the calculated fractions of stopping power as a function of proton energy in which contributions from both proton and neutral hydrogen interactions are included. The term 'ion-potential' includes both the energies used to overcome the ionization potential of water molecules and those less than the cut-off of 1 eV for ejected electrons by ionization and electron loss. The fractions of 'e-capture' include energies with less than the cut-off for electrons ejected by electron loss of neutral hydrogen. Energetic electrons ejected by both ionization and electron loss processes are lumped under the component of '2ndry electrons'. Above 20 keV the kinetic energy transferred to ejected secondary electrons is the largest part of the stopping power. The fraction of $0.67 > 300$ keV agrees well with the

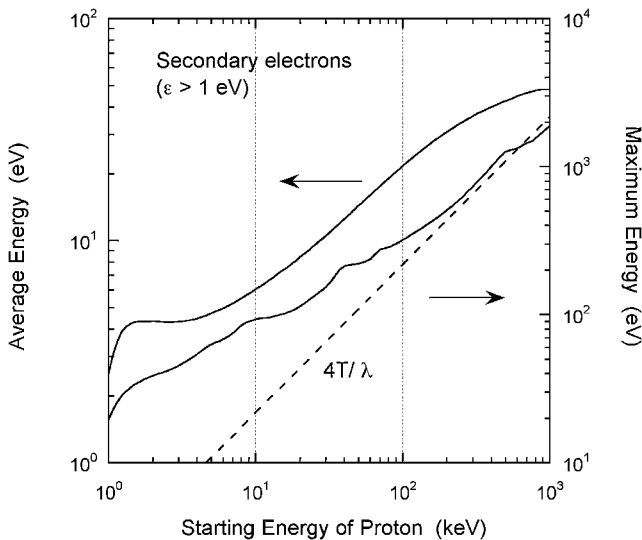


Figure 10. Average energy and maximum energy of secondary electrons generated by the slowing down of protons as a function of the starting energy of proton. The maximum energy $4T/\lambda$ obtained from the kinematical relationship is in proportion to the initial proton energy T as shown by the straight dashed line.

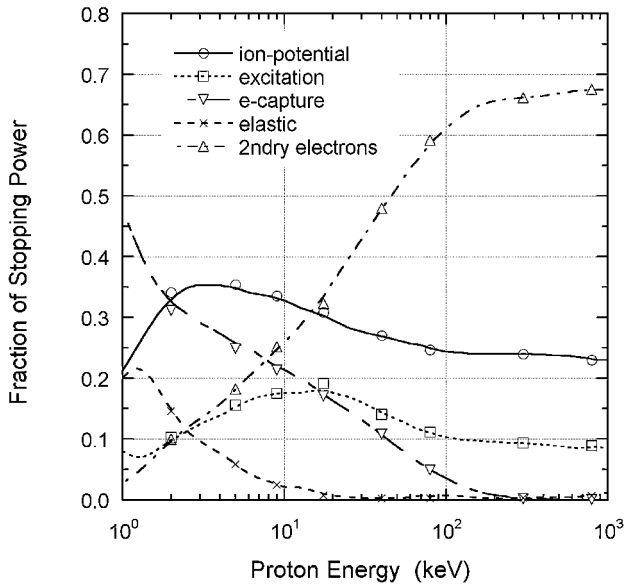


Figure 11. Fractions of stopping power for water vapour for protons as a function of proton energy. The fraction of ‘ion-potential’ is the kinetic energy used to overcome the ionization potential of water and ‘2ndry electrons’ is the kinetic energy of ejected electrons due to ionization and electron loss processes of neutral hydrogen.

experimental fraction 0.67 ± 0.04 for 1 MeV protons, which has been estimated by Toburen and Wilson (1977) and Lynch *et al.* (1976). Charge exchanges and nuclear elastic scattering become significant < 10 keV.

5.2. Radial dose distribution

The radial dose distribution, which is the energy absorbed locally at a certain radial distance from the path of a primary particle, has been used to check the reliability of track structure calculations. Figure 12 compares various data on the radial dose profiles around the path of 1 MeV proton tracks. In this case, the energy depositions along the proton track itself, i.e. ‘ion-potential’, ‘excitation’, ‘electron capture’ and ‘elastic’ denoted in figure 11, are excluded in dose calculations. Energies deposited only due to secondary electrons were accumulated within a coaxial shell with a radial interval of 0.5 nm along the proton track from the origin to $2 \mu\text{m}$. The total energy deposited in this interval of the proton range is 50 keV. Data are compared with two sets of experimental data (Wingate and Baum 1976) and calculated data of Butts and Katz (1967), Cucinotta *et al.* (1999) and the code *pits* (1999). Reasonable agreement is observed over a wide range. At larger ranges the statistics of the track is low as only 50 tracks were analysed as each track approximately requires ~ 10 Mb of disk space.

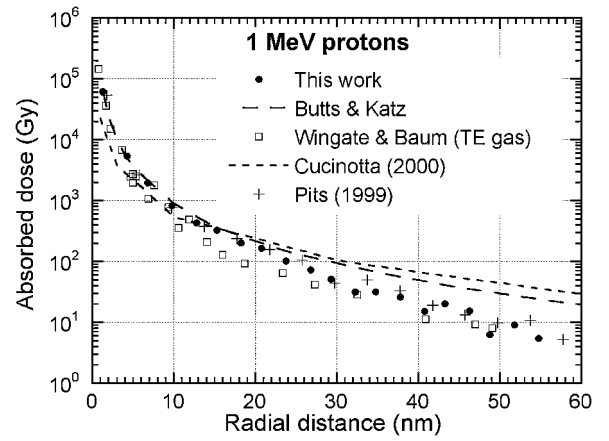


Figure 12. Calculated radial dose profiles at ~ 1 MeV proton paths compared with experimental, calculated and track structure data.

5.3. Restricted stopping powers

Energy restricted stopping power (ICRU 1970), L_Δ , is defined as that part of the total energy loss dE/dl which is associated with excitation-ionization energy transfers up to a cut-off value Δ . Radius restricted stopping power, L_r , is defined as that part of the total energy loss dE/dl which is deposited within a cylinder of radius r and length dl , centred along the particle track. These are useful for checking the reliability of track structure calculations. Figure 13 shows the calculated ratio of L_Δ to L_∞ for the proton energies of 0.1, 0.2 and 1 MeV as a function of cut-off energy Δ , where the symbol L_8 means the unrestricted stopping power and is used when all possible energy transfers are included. The track length dl was set at 50, 120 and 2000 nm from the origin for $T = 0.1, 0.2$ and 1 MeV respectively in which the energy deposited amounts to 5% of the

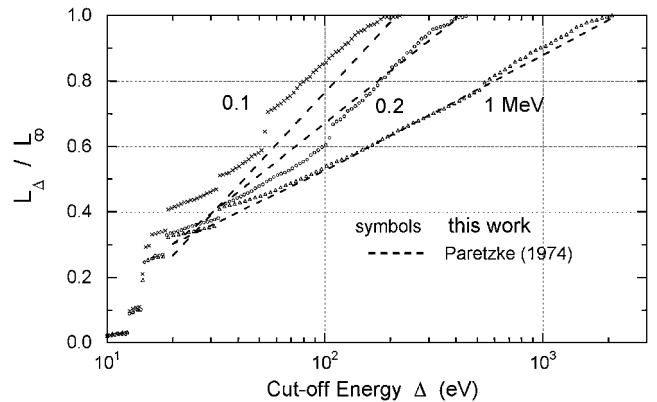


Figure 13. Ratio of energy restricted stopping power L_Δ to unrestricted stopping power L_∞ for the proton energies of 0.1, 0.2 and 1 MeV as a function of cut-off energy Δ . Calculated data (dashed lines) are from Paretzke (1978, 1987).

primary energy of the proton. Discontinuities at $\Delta = 13, 15, 19$ and 32 eV in the curves correspond to ionization thresholds for each orbital of water. Additional discontinuities caused by electrons stripped in electron-loss process are found at $\sim T/\lambda$. These are 54 and 108 eV for $T=0.1$ and 0.2 MeV respectively. For 1 MeV protons nearly 50% of the stopping power is contained in energy transfers < 100 eV. Dashed lines present the calculated values from Paretzke (1978, 1987), in which the discontinuities are not generated because calculations do not take into account the molecular structure. Agreement between present Monte Carlo calculations and analytical calculations becomes better at higher proton energies.

Whereas L_Δ can readily be calculated from basic differential cross-sections by simple integration, it can not be measured directly; radius restricted stopping power L_r can be measured directly but not readily calculated since full track structure calculations are needed. Figure 14 shows the ratio of L_r to unrestricted stopping power L_∞ as a function of radial distance from the proton path having primary energies of $0.1, 0.2$ and 1 MeV. The present calculation for 1 MeV protons is in reasonable agreement with the experimental data for the tissue equivalent gas (Wingate and Baum 1976). For comparison the calculated values of Paretzke (1978, 1987) are shown for 0.2 and 1 MeV track segments. A large fraction of the energy lost by fast ions stays within the first nanometer around the path, but with increasing ion

energy a large fraction can be transported by energetic electrons to large distances from that path.

5.4. Event size

Calculations of distributions for energy imparted by protons in nanometer sites were performed. The definition of the energy imparted, to the matter in a volume is $\varepsilon = \sum \varepsilon_i$, where the summation is performed over all energy deposits, ε_i , in that volume in a single interaction, ε_i (ICRU 1983). ε_i is the radiation energy converted to forms other than kinetic energy of ionizing particles, i.e. binding energy of stripped-away electrons, excitation energy for electronic excitation, etc. The data of ε_i is available from the event-by-event Monte Carlo simulation. Both ε_i and the energy imparted, ε , are the stochastic quantities. Figure 15 shows representative examples of frequency distributions, $f(\varepsilon)d$, for energy imparted by 100 keV protons passing along a diameter through spherical sites of diameters ranging from 1 to 50 nm. The data were obtained from the analysis of 100 proton tracks. The histograms are for frequency of event size, where event size is measured by the number of energy deposition produced within the site boundary by single proton tracks. For the smaller sites, the distributions look essentially like simple Poisson distributions in spite of the statistical fluctuation. For the larger sites, the distributions exhibit the skewed Gaussian shape similar to energy loss straggling distributions.

6. Conclusions

In this report we have described the development of a Monte Carlo track structure code for protons in the energy range 1 keV to 1 MeV. The code *lephist* provides coordinates of all molecular interactions, amount of energy deposited along the track and the type of interaction at each event for initial proton energies 1 keV to 1 MeV. In spite of several assumptions in compiling the basic data especially for neutral hydrogen interaction, the results such as fractional stopping powers, ranges, W values and microdosimetric analysis provide reasonable and good agreement with the experimental data and other theoretical data. Our data on stopping cross-section is higher than the published ICRU data because we have avoided any adjustment of parameters for various processes. This discrepancy may arise from the two different method used in ICRU and ours to calculate the stopping power data. ICRU stopping power data have been calculated using the effective charge method while we use the experimental cross-sections and the mean energy transfer. We could have used an alternate approach by starting with a

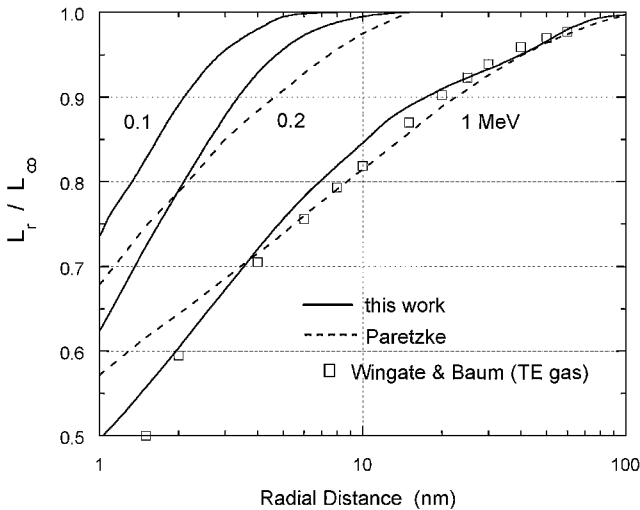


Figure 14. Ratio of radius restricted stopping power L_r to unrestricted stopping power L_∞ for the proton energies of $0.1, 0.2$ and 1 MeV as a function of radial distance from the proton path in water. Dashed curves are calculated values from Paretzke (1978, 1987). The experimental data of Wingate and Baum (1976) were obtained for protons in tissue equivalent gas.

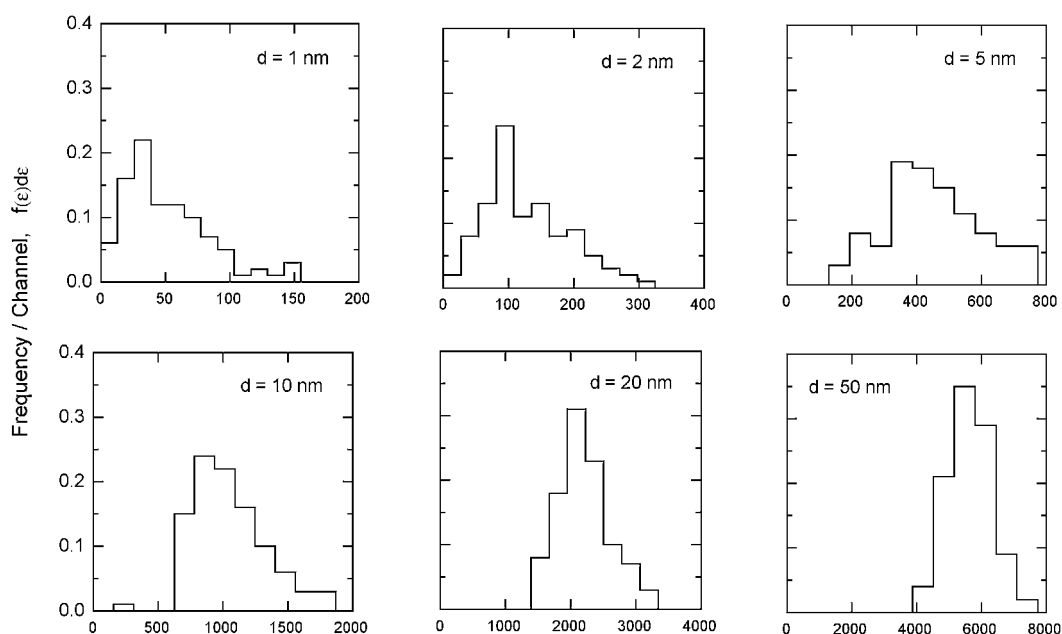


Figure 15. Examples of frequency distributions $f(\epsilon)d$ for energy imparted by 100 keV protons passing diametrically through 1–50-nm diameter spherical sites in unit density water. Spectra were from the trial of 100 protons.

fit to the ICRU stopping power data and adjusted all parameters accordingly in calculating the cross-sections. In summary, we have presented cross-sections, single and doubly differential, and stopping power data needed for simulation of full slowing down track of low energy protons in the range 1 keV to 1 MeV.

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